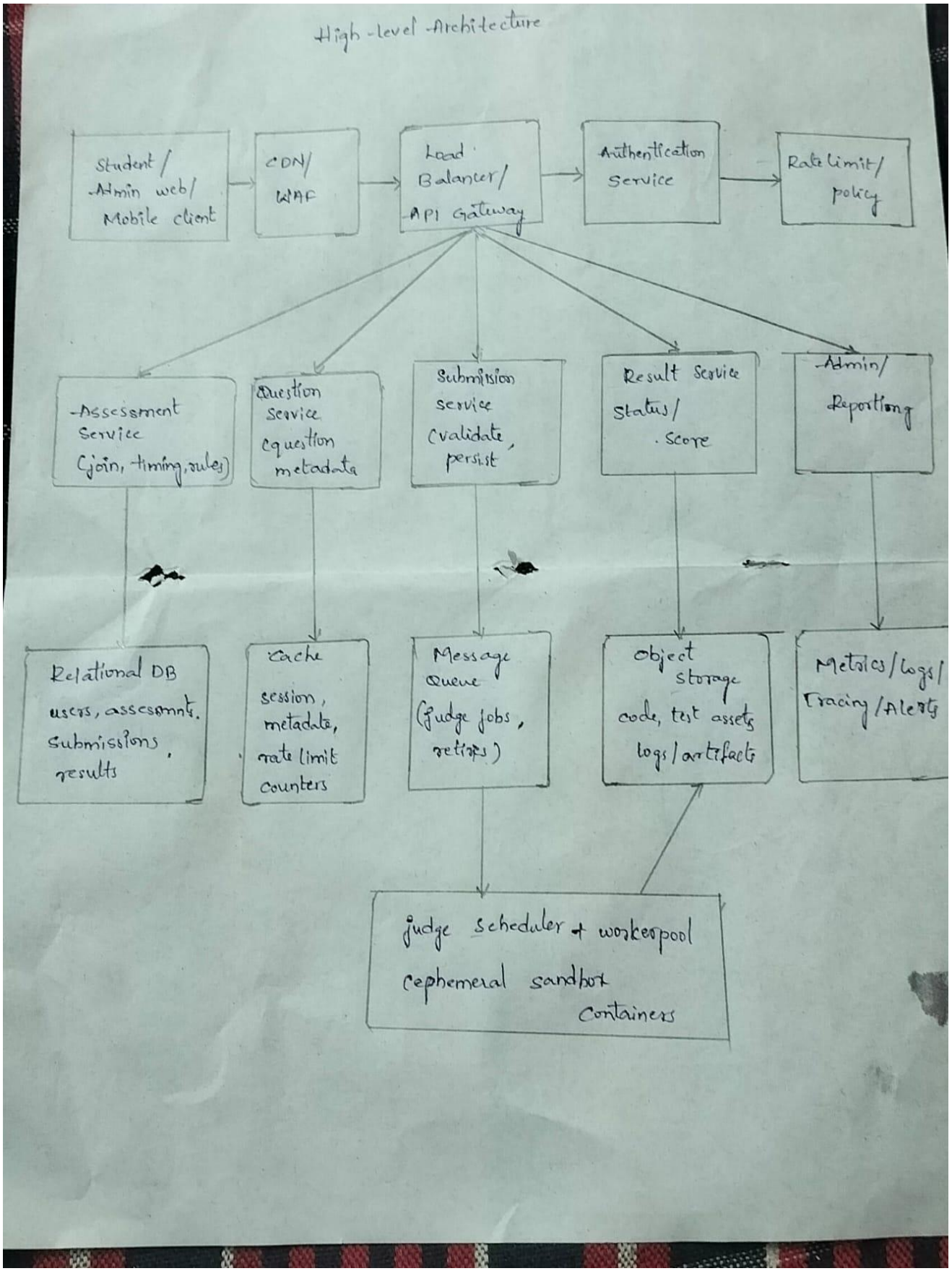

High-level Architecture
Student / Admin web / Mobile client
CDN / WAF
Load Balancer / API Gateway
Authentication Service
Rate limit / policy
Assessment Service (join, timing, rules)
Question service (question metadata
Submission service (validate, persist
Result Service Status / Score
Admin / Reporting
Relational DB users, assessmnt. submissions, results
Cache session, metadata, rate limit counters
Message Queue (Judge jobs, retries)
object storage code, test assets logs / artifacts
Metrics / logs / Tracing / Alerts
judge scheduler + workerpool
Cephemeral sandbox Containers